

SPIRAL MODEL: RISK-DRIVEN APPROACH

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Abstract - The Spiral Model is a software development methodology that combines elements of design and prototyping in stages. Created to address the limitations of traditional linear approaches, it focuses on managing risks and iterative development, making it particularly suitable for complex and high-risk projects. This paper explores the Spiral Model's structure, benefits, and applications, highlighting its emphasis on flexibility and early risk detection.

Keywords – Spiral Model, Risk-Driven Approach, Software Development Methodology, Pangulu Nagori

I. INTRODUCTION

Software engineering is a systematic and structured field that covers all important parts of software production, starting from the very first stage of system specifications to ongoing maintenance. It plays a key role in today's technology, focusing on delivering high-quality software that works well and on time [1].

The Spiral Model is known for its flexible and risk-focused approach. Unlike the traditional waterfall method, this model works well for projects that need frequent changes. It's particularly useful in areas like mobile app development, where requirements can shift quickly. The Spiral Model allows for adjustment and emphasizes continuous improvement, making it a helpful choice for developers dealing with changing needs.

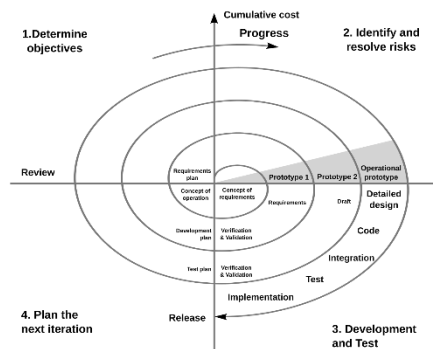


Fig. 1 The Spiral Model

II. BACKGROUND AND HISTORY

1986: Barry Boehm introduces the Spiral Model, describing it as a four-phase cycle. Each phase—Planning, Risk Analysis, Engineering, and Evaluation—completes in each "spiral" or iteration. Boehm's model was among the first to combine iterative development with risk management, and it highlights the importance of early risk identification along with flexible adaptation [2].

1990s: The model finds adoption in high risk industries like aerospace and defense, where software requirements change frequently. This flexibility helps teams address risks early and make adjustments as new issues or project needs come up [3]. The Spiral Model quickly became recognized as a useful framework for managing complex projects.

2000s: Software development teams start adapting the Spiral Model to work with Agile methods. Its risk-focused and iterative nature complements Agile's flexibility, resulting in hybrid development practices in areas that need ongoing improvement and responsiveness [4].

Today: The Spiral Model continues to be influential, particularly in projects needing both flexibility and solid risk management. It's commonly used in areas like medical software, financial systems, and critical engineering. The model's structured but adaptable nature makes it a preferred method for projects that benefit from thorough risk assessment and iterative progress [5].

III. APPLICATION OF SPIRAL METHODOLOGY

One of the government units in Indonesia, Pangulu Nagori Amborokan Panei Raya Office were planning to design and develop an employee attendance application based on mobile to record their workers' attendance []. The issues arise when this organization still uses traditional methods to track their employee attendance by using an attendance book which becomes concerning about security and management issues [6].

The spiral model was employed in this software development. The activities were divided into different stages that were suitable for each one of them. The activities that being employed in the development were listed and explained below [6]:

A. Customer Communication

Customer Interviews were conducted to gather all necessary information about the organization's operational operation.

1) *Problem Analysis:* The Pangulu Nagori Amborokan Panei Raya Office currently uses a manual attendance book to log employee attendance. This manual system makes it hard to manage attendance data efficiently and takes a lot of time and effort. It also creates issues in quickly accessing and tracking employee attendance records.

2) *System Requirements Analysis:* To improve this, the office plans to introduce an Android-based app. This app would help both staff and administrators log and monitor attendance more easily.

B. Planning

At this stage, the organization determines the requirements to perform research, such as location, time, tools and materials.

1) *Research Sites:* The project will be conducted at the Pangulu Nagori Amborokan Panei Raya Office, located on Jalan Besar Panei Raya in North Sumatra.

2) *Research Tools:* This study used several devices and software. The main devices included an ASUS Zenfone Android phone, an Acer Aspire laptop with an Intel Core i5 processor, NVIDIA graphics, and 4GB RAM, plus a Max Pro 1 device with 6GB RAM, 16GB storage, and an Adreno GPU. Key software tools were Windows 10 Pro, Microsoft Word for writing, Kodular for app development, Airtable for data management, Visio for diagrams, Google Chrome, and Android 9.1.

3) *Research Materials:* Data came from resources provided by the Pangulu Nagori Amborokan Panei Raya Office, including attendance logs, participation reports, and institutional guidelines for tracking attendance.

C. Analysis Risk

The evaluation of functional requirements of the product was conducted first before analyzing the risk of the intended product. This includes login page, options menu, employees menu, attendance menu and report menu.

D. Engineering

The detailed specifications were constructed in this phase, including system architecture, materials requirements and aesthetics of the system. UML was involved in this phase to design the structure and attributes of the system.

Multiple diagrams were developed to handle different aspects of the system such as interaction between the users

and the system, workflow of the system and classes of attributes and their connections.

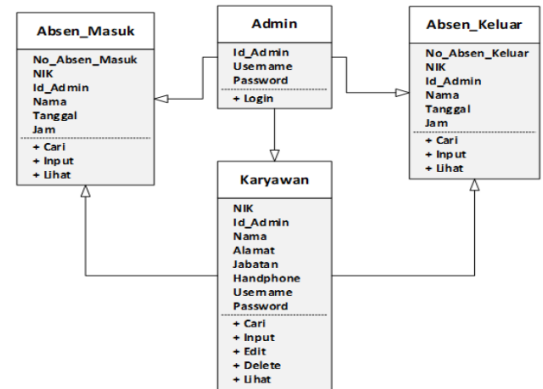


Fig. 2 Class diagram for the employee attendance application system

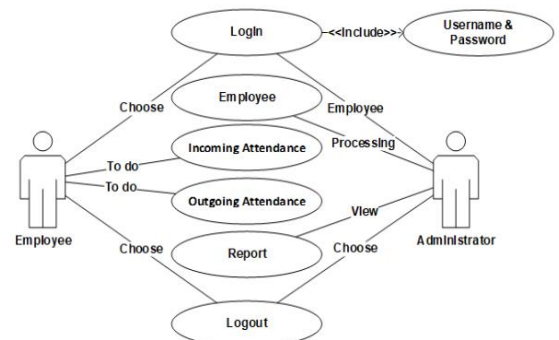


Fig. 3 Use case diagram for the employee attendance application system

E. Construction & Release

During this phase, the implementation of employee attendance application was put into work. All the features of the application were being developed into a mobile application. There are 5 modules that were employed in this application including:

1) *Login Page:* Provide security for the system, user must log in before using the application.



Fig. 4 Example of login page of the application

2) *Attendance Page*: For employees to key in their attendance either check in or check out.

3) *Options menu page*: Have 3 options for administrator can choose which are employee, report and logout menu.

4) *Employee Page*: For administrator to process the employee's data.

5) *Report Page*: Administrator can view the summary of employees' attendance throughout the month.

F. Customer Evaluation

A questionnaire was developed to evaluate the application that is just being built, focusing on usability, design and performance for the employee attendance process. The questionnaire was distributed to ten employees at Pangulu Nagori Amborokan Panei Raya Office to gather their feedback on the application.

IV. COMPARISON OF SPIRAL METHODOLOGY

Barry Boehm, who originally developed the Spiral Model, highlights its structure and adaptability. He presents the Spiral Model as a cycle with stages—planning, risk analysis, engineering, and evaluation—designed to repeat in a loop. This looping structure helps manage risk by continuously evaluating and adjusting the project as it progresses. Boehm argues that the model's focus on risk assessment makes it particularly suited for complex projects with high uncertainties.

In contrast, the second study, a more recent one by Sari et al., examines the Spiral Model's performance in real-world, high-risk environments like aerospace. Sari et al. find that the model's iterative, adaptable approach works well in these fields because it can handle constant changes in requirements, ensuring that the software adapts as the project evolves. They view the model as valuable in risky environments but acknowledge it may be resource-intensive.

In conclusion, both studies suggest that the Spiral Model is effective, especially for high-stakes projects. Boehm's foundational perspective on its structure is invaluable, but Sari et al. provide useful insights into its modern applications. Overall, Boehm's work is crucial for understanding the core theory, while Sari et al. offer a practical evaluation. Thus, Boehm's foundational study might be considered slightly stronger in understanding the model's design, although Sari et al. effectively highlights its value in current industry practices.

V. CONCLUSION

In conclusion, the Spiral Model is a great approach for handling the challenges of software development, especially when flexibility and managing risks are important. Its iterative process allowed the developers team to continuously improve the project, addressing issues as they came up and adapting to changing needs [7]. By involving stakeholders throughout the different phases, the team could ensure that the final product aligned with user expectations. The focus on early risk detection also meant that potential problems could be managed before they became bigger issues. Overall, the Spiral Model's capability to adapt makes it a strong choice for projects that need to stay responsive to changes.

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